

|   |     |                                                                 |                                                                                                                                       |                |     |     |
|---|-----|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|-----|
| 2 | (a) | (i)                                                             | horizontal speed constant at $8.2 \text{ m s}^{-1}$<br>vertical component of speed $= 8.2 \tan 60^\circ$<br>$= 14.2 \text{ m s}^{-1}$ | C1<br>M1<br>A0 | [2] |     |
|   |     | (ii)                                                            | $14.2^2 = 2 \times 9.8 \times h$ (using $g = 10$ then $-1$ )<br>vertical distance $= 10.3 \text{ m}$                                  | C1<br>A1       | [2] |     |
|   |     | (iii)                                                           | time of descent $= 14.2 / 9.8 = 1.45 \text{ s}$<br>$x = 1.45 \times 8.2$<br>$= 11.9 \text{ m}$                                        | C1<br>A1       | [2] |     |
|   | (b) | (i)                                                             | smooth path curved and above given path<br>hits ground at more acute angle                                                            | M1<br>A1       | [2] |     |
|   |     | (ii)                                                            | smooth path curved and below given path<br>hits ground at steeper angle                                                               | M1<br>A1       | [2] |     |
| 3 | (a) | force $=$ rate of change of momentum (allow symbols if defined) |                                                                                                                                       |                | B1  | [1] |